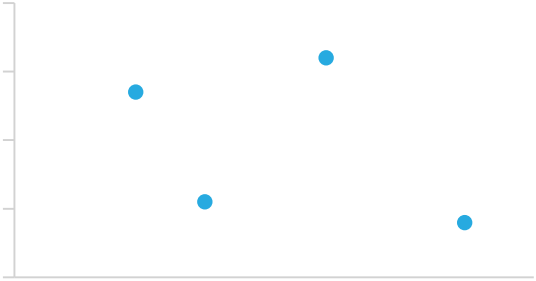
THE SCIENCE OF FACT-BASED PERSUASION BASIC GRAPHS

Michael Smith
mike@slideheroes.com

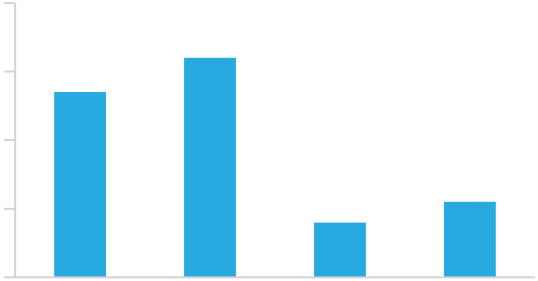
SLIDEHEROES

REPRESENTING QUANTITATIVE VALUES

POINTS

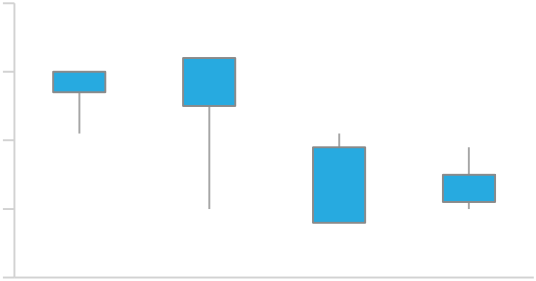


BARS

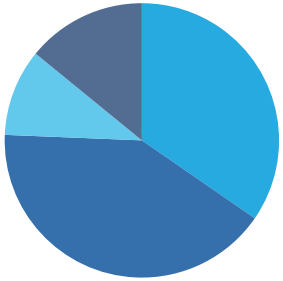


LINES

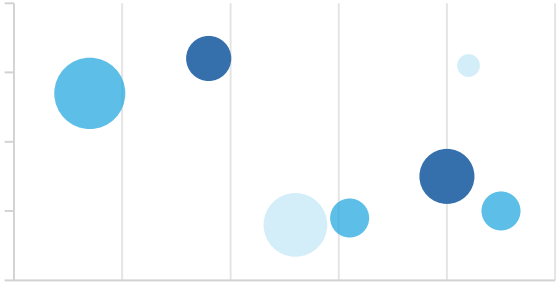
BOXES



SHAPES WITH 2D AREAS



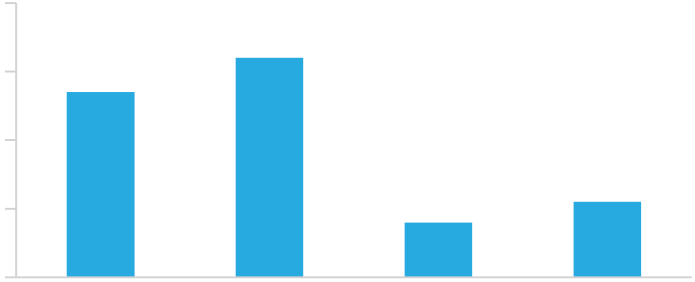
VARYING COLOR INTENSITIES



BASIC CHART TYPES

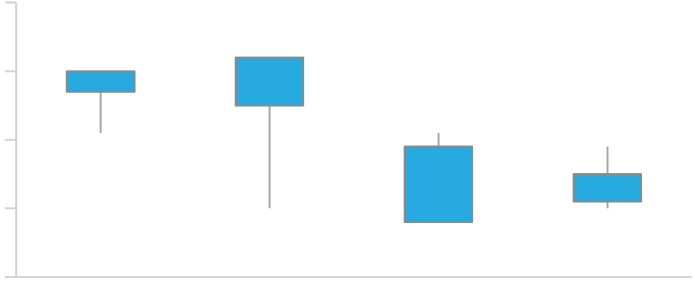
POINTS

BARS



LINES

BOXES



8 TYPES OF RELATIONSHIPS

NOMINAL COMPARISON

TIME SERIES

RANKING

PART-TO-WHOLE

DEVIATION

DISTRIBUTION

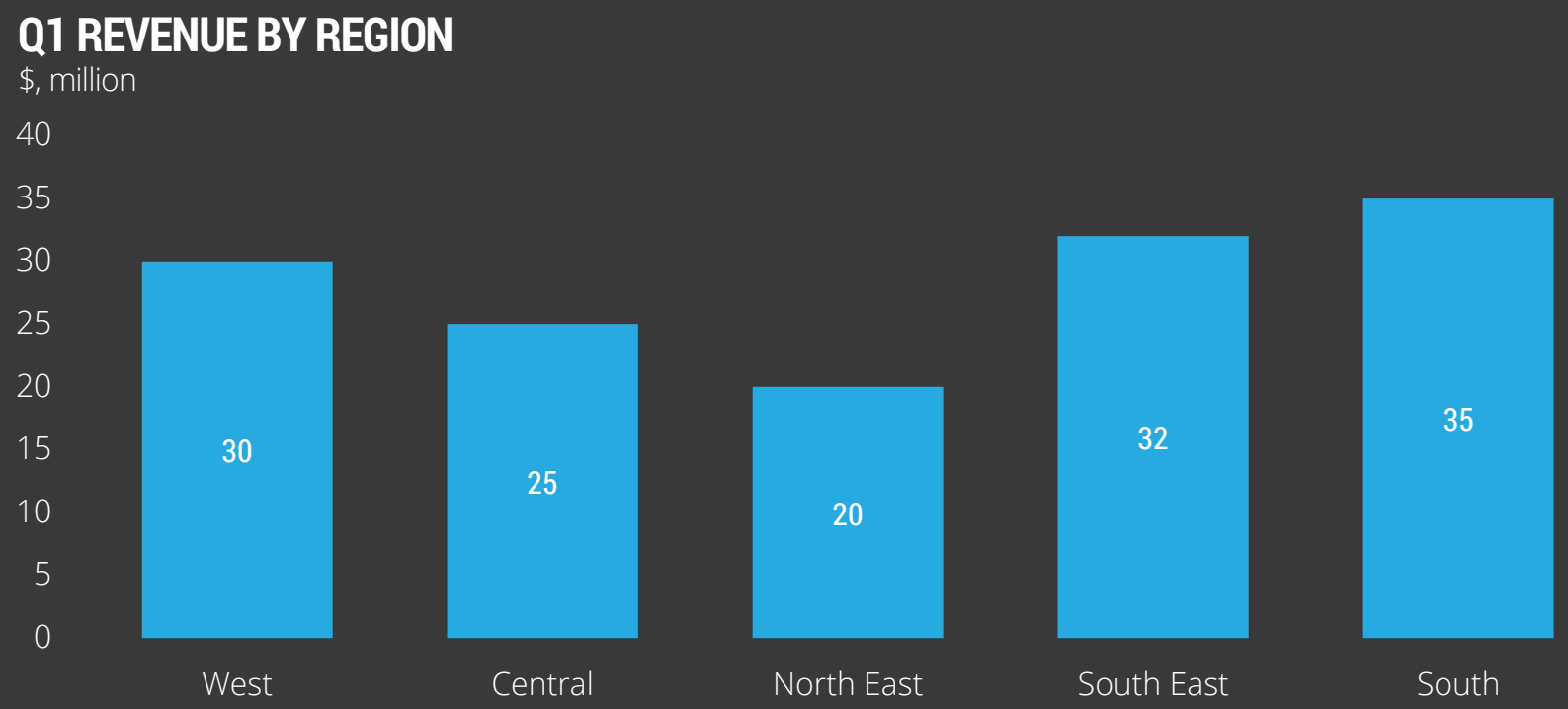
CORRELATION

GEOSPATIAL

Basic Graphs

1. NOMINAL COMPARISON

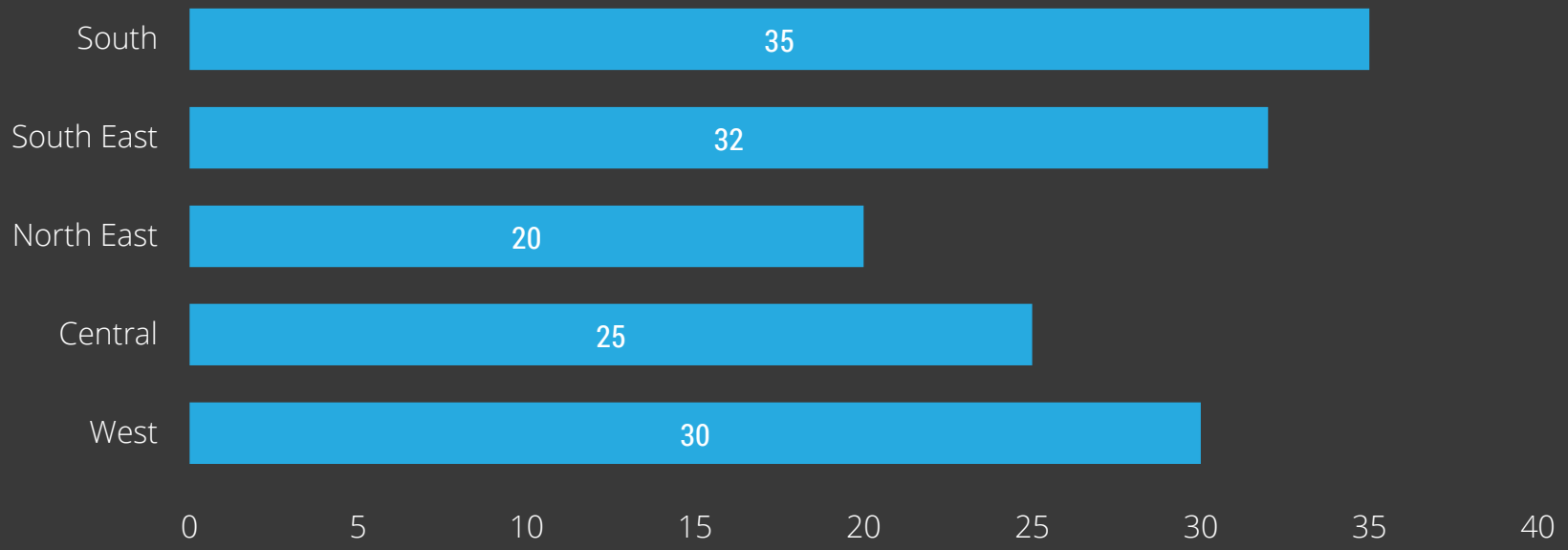
NOMINAL COMPARISON BAR CHARTS



NOMINAL COMPARISON BAR CHARTS

Q1 REVENUE BY REGION

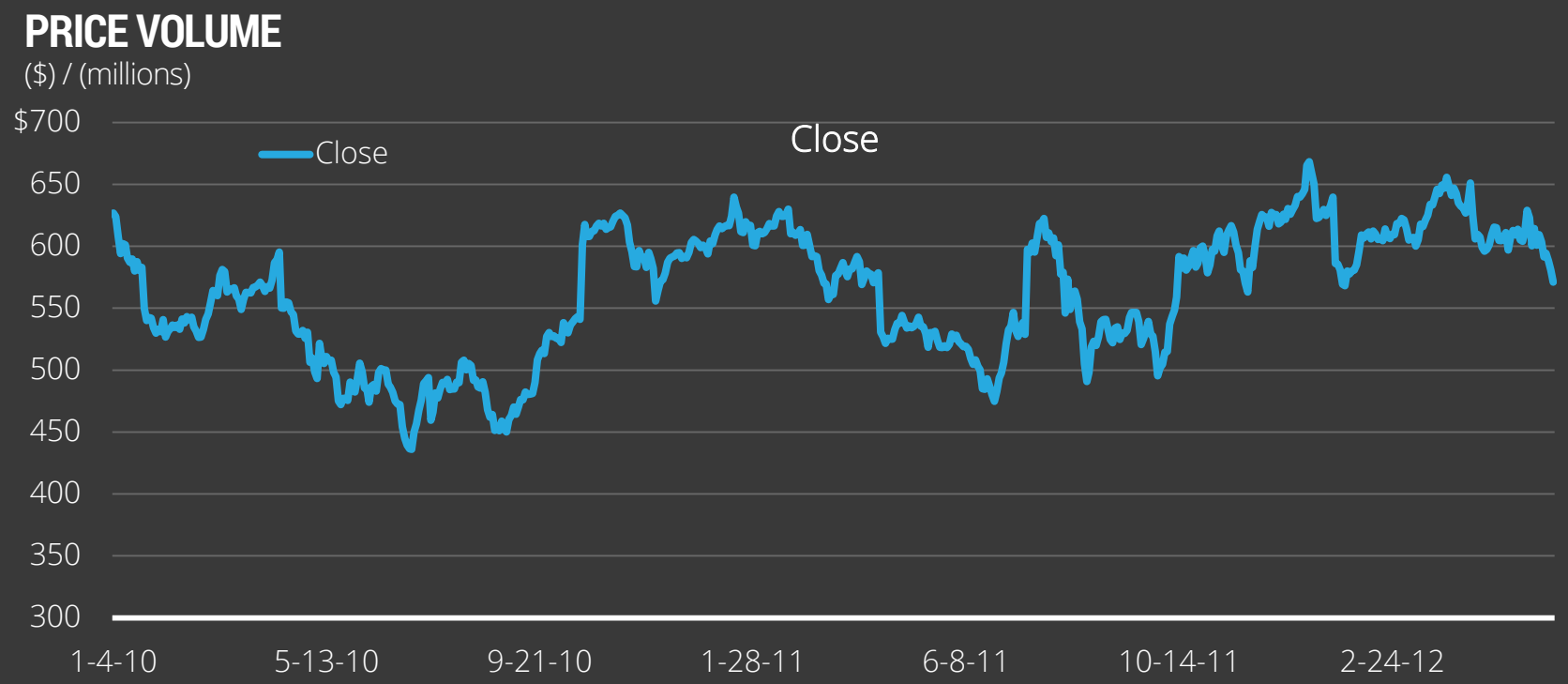
$, million



Basic Graphs

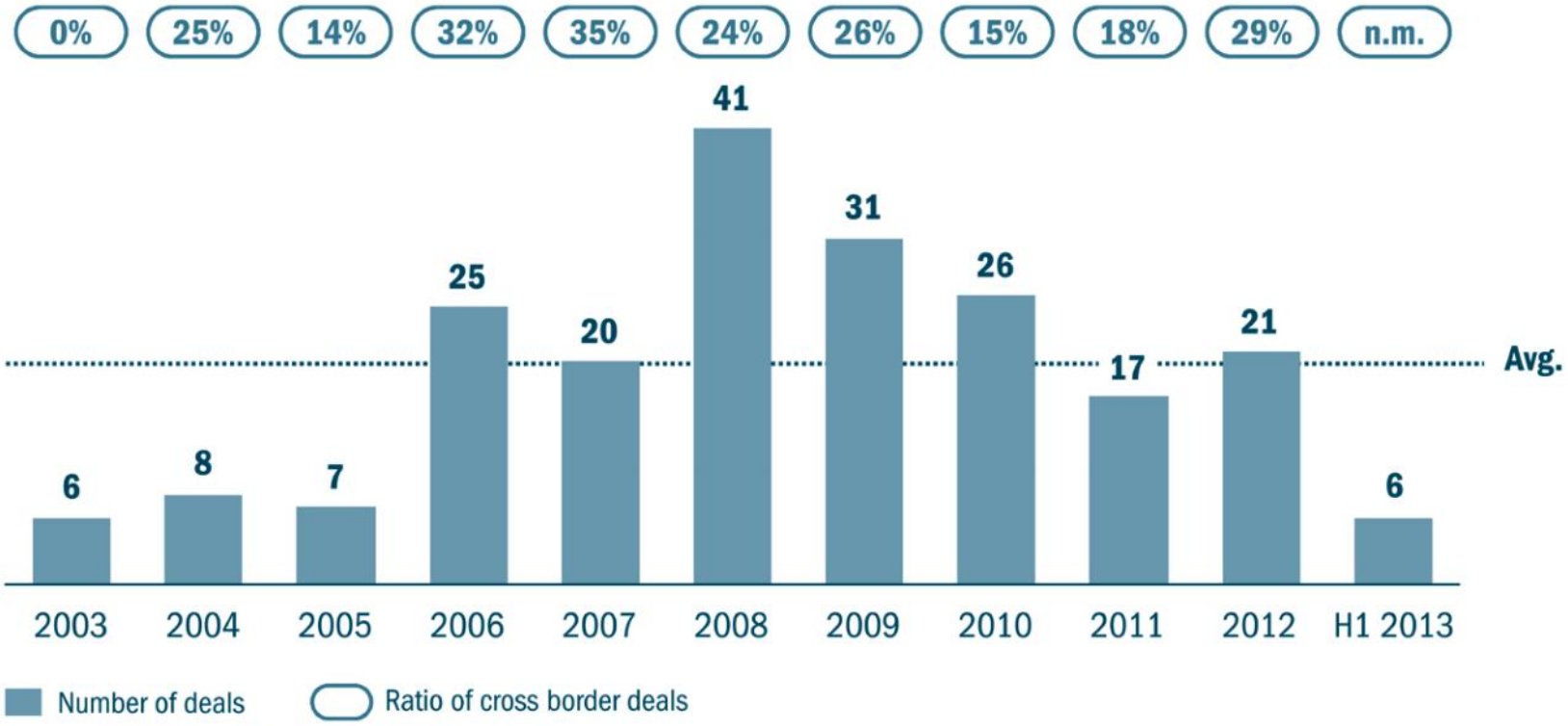
2. TIME SERIES

HISTORICAL PRICE/VOLUME



SOURCE: Yahoo finance

Figure 7: Number of deals in the private banking industry in Europe

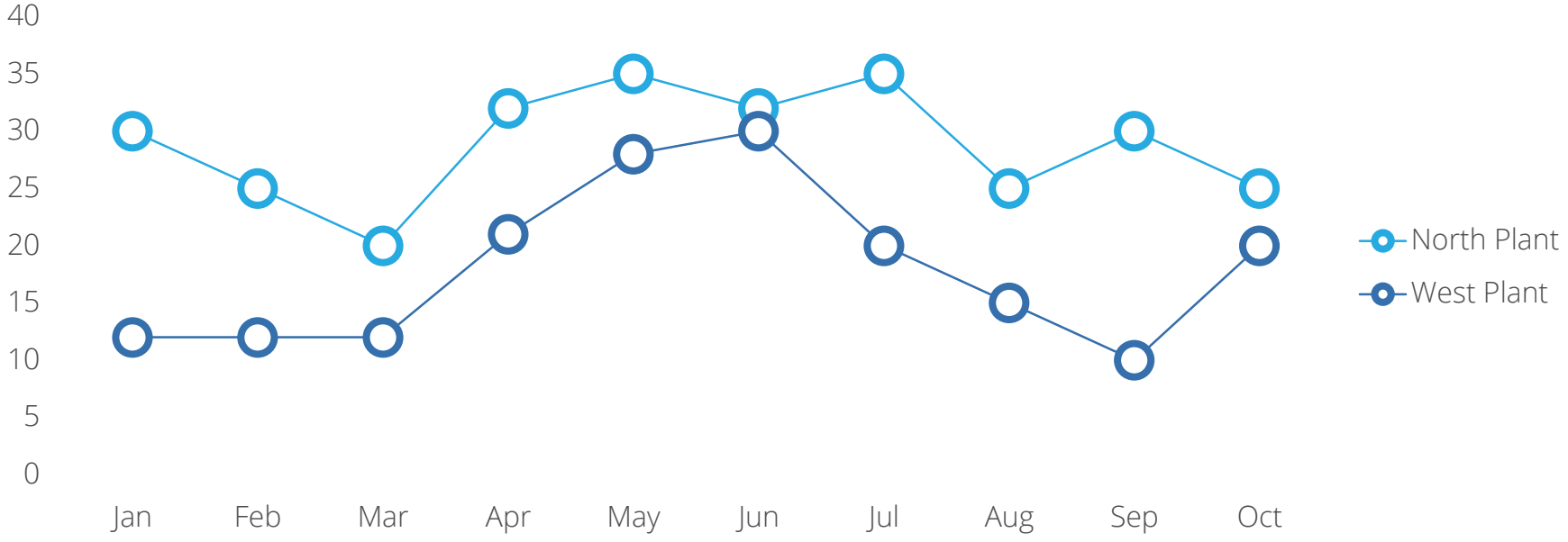


Source: SNB; Roland Berger Strategy Consultants

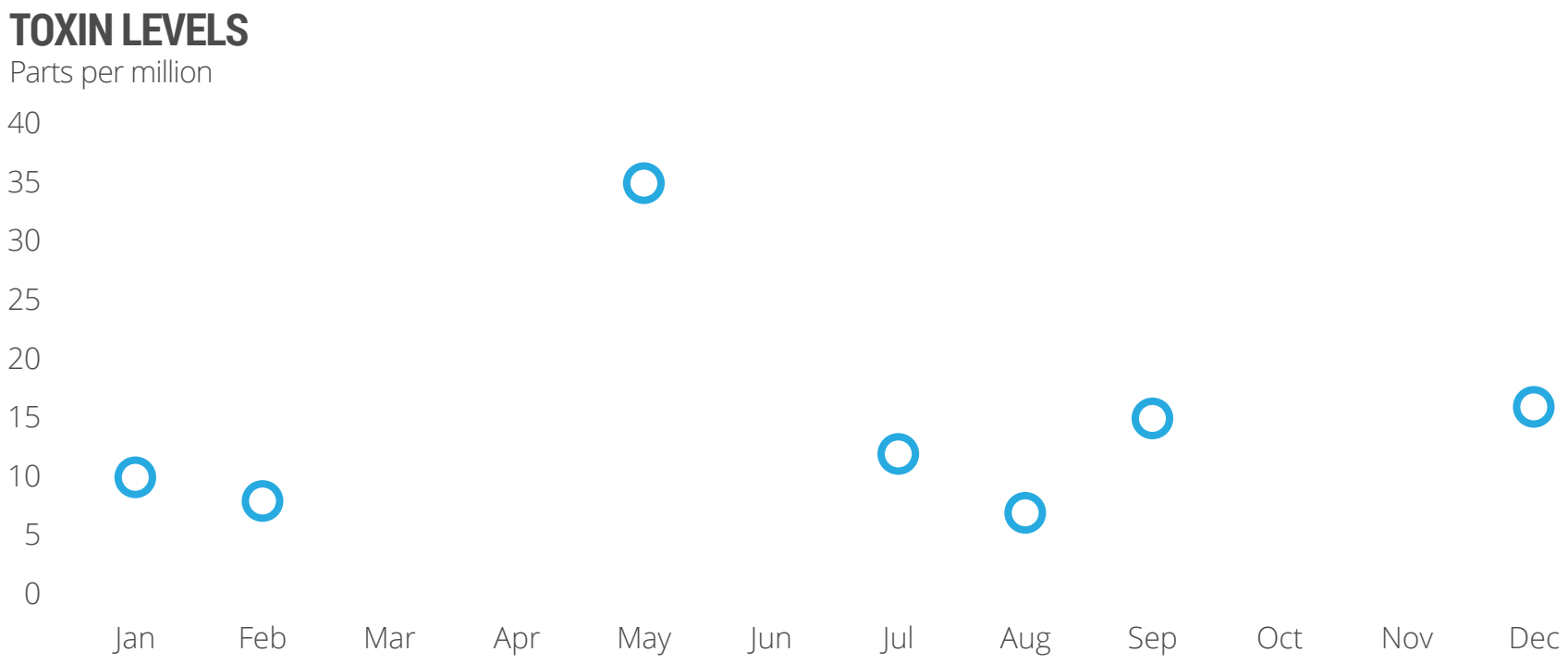
LINE GRAPH TIME SERIES

MANUFACTURING UNITS

Monthly, million



TIME SERIES AS POINTS



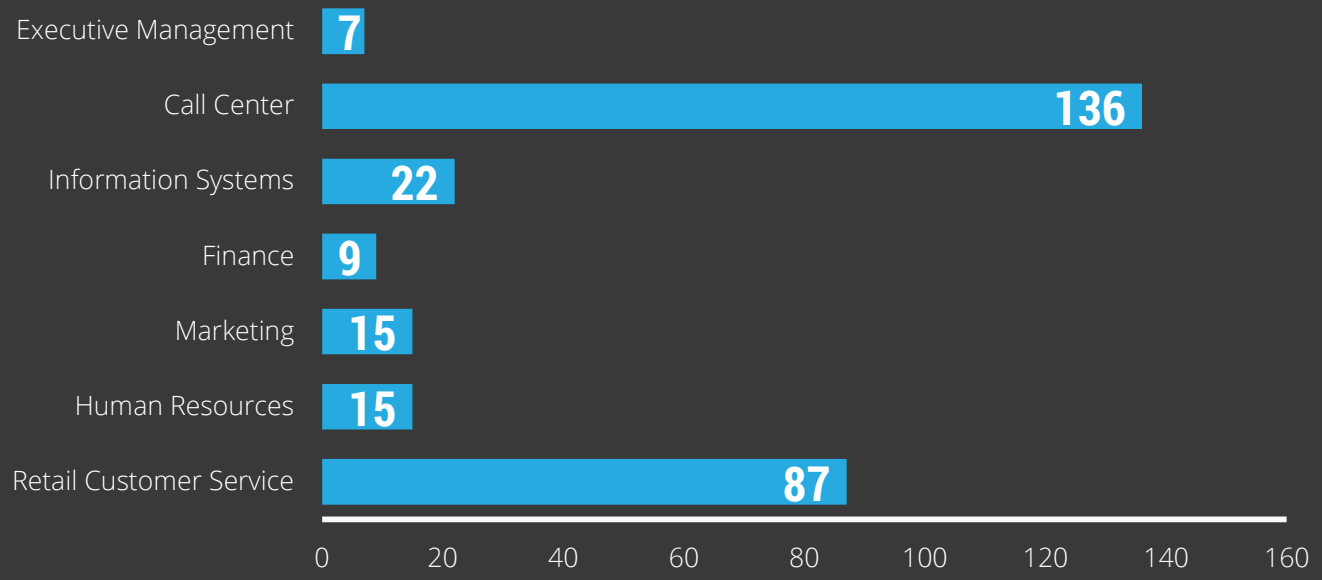
Basic Graphs

3. RANKING

RANKING HORIZONTAL BAR CHARTS

NUMBER OF EMPLOYEES BY FUNCTION

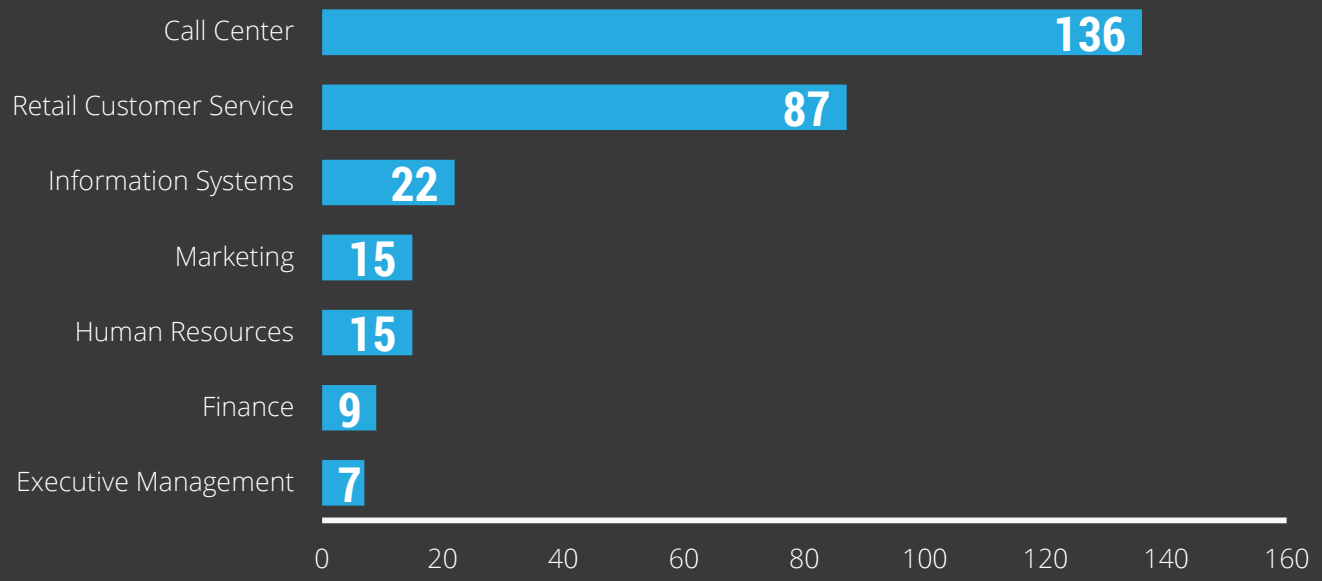
Full-time equivalents, January 2014



RANKING HORIZONTAL BAR CHARTS

NUMBER OF EMPLOYEES BY FUNCTION

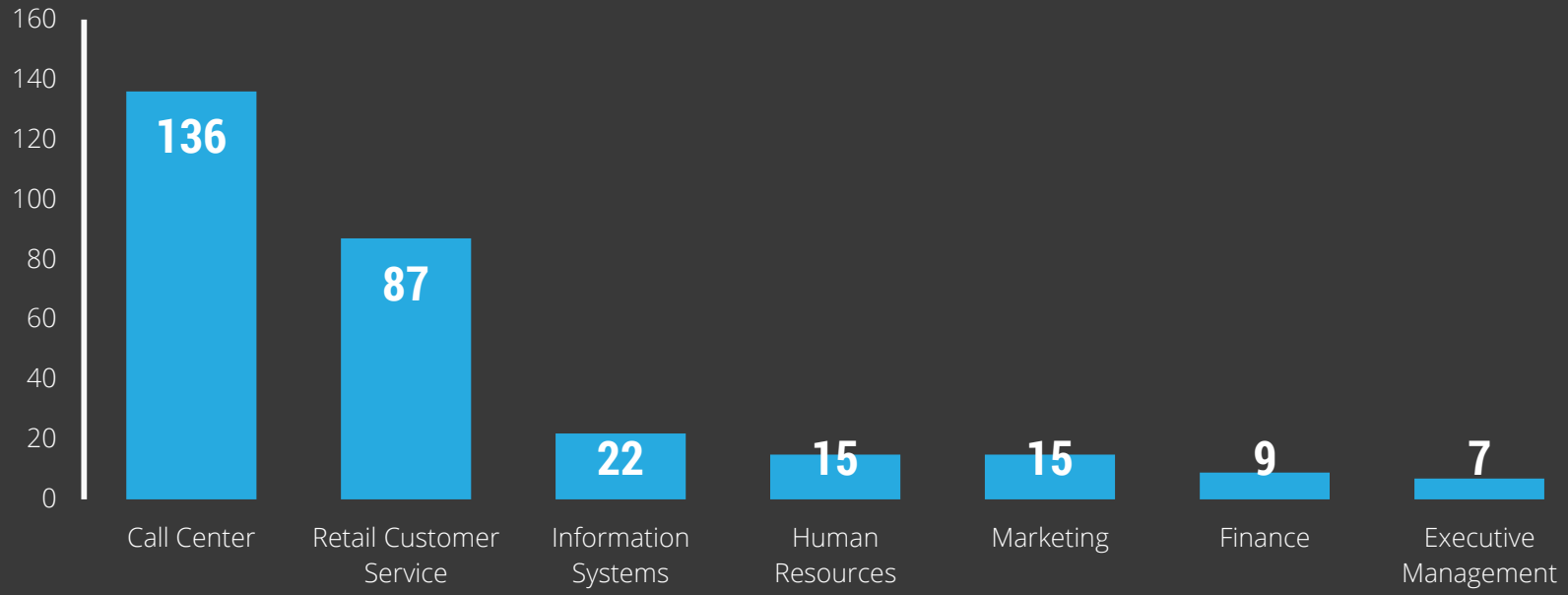
Full-time equivalents, January 2014



RANKING HORIZONTAL BAR CHARTS

NUMBER OF EMPLOYEES BY FUNCTION

Full-time equivalents, January 2014



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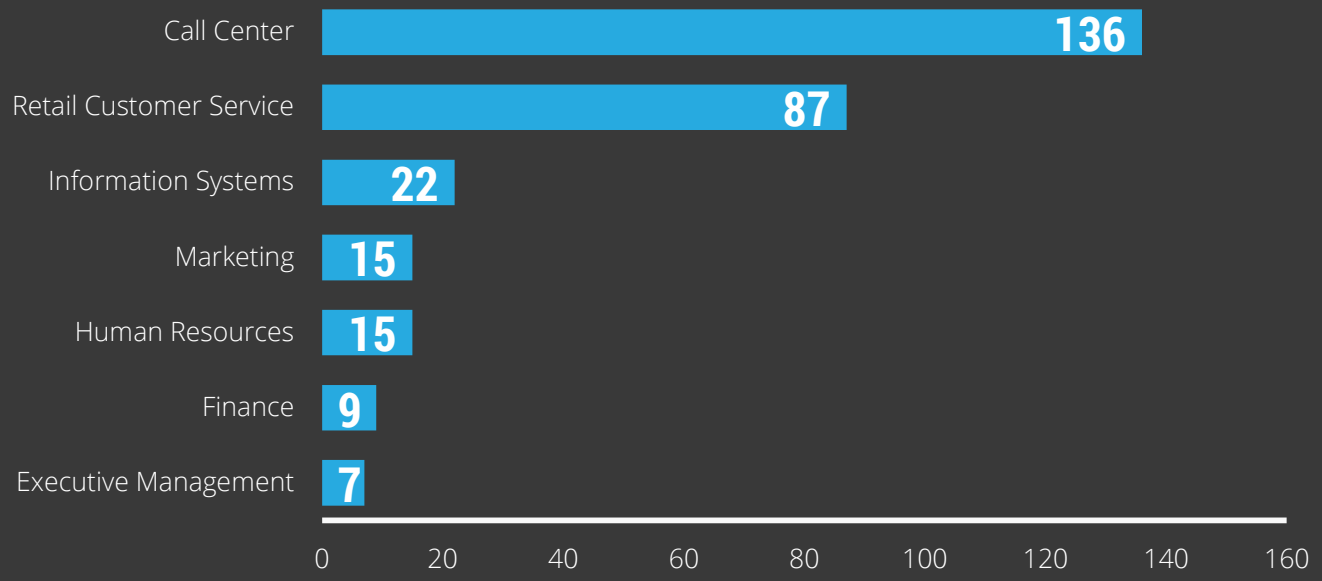

RANKING HORIZONTAL BAR CHARTS

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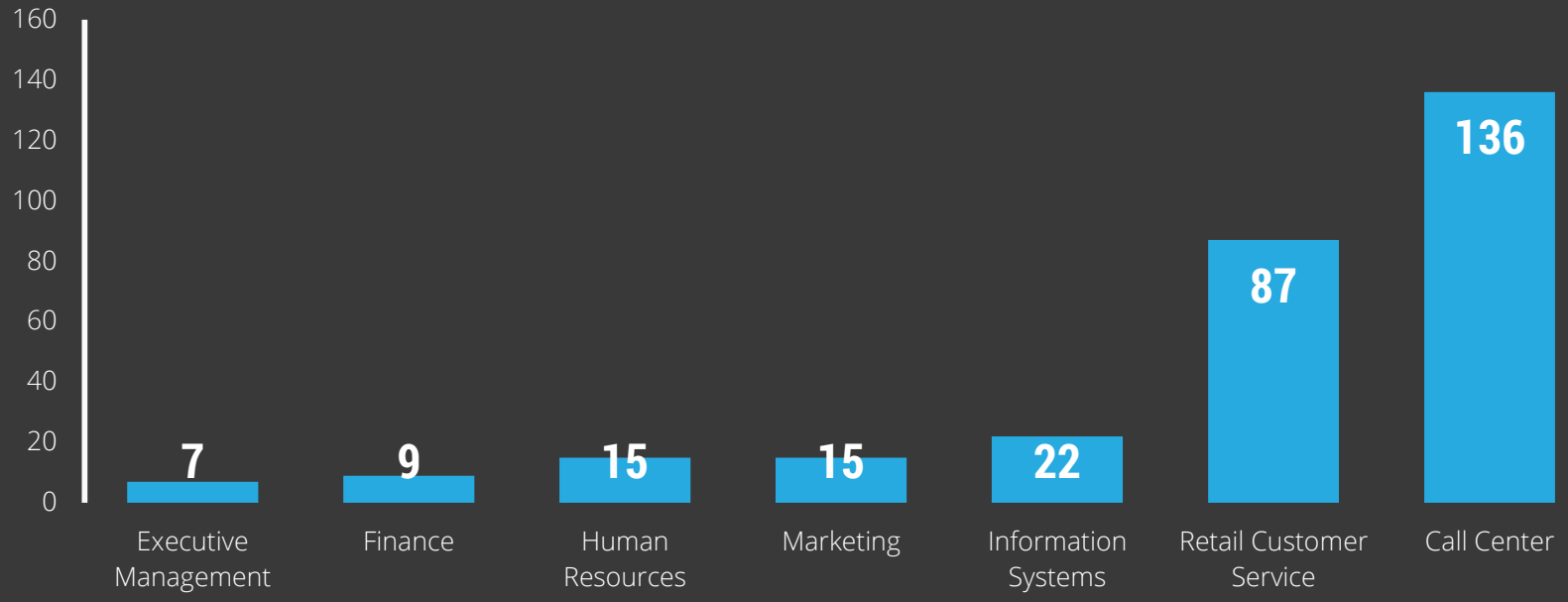
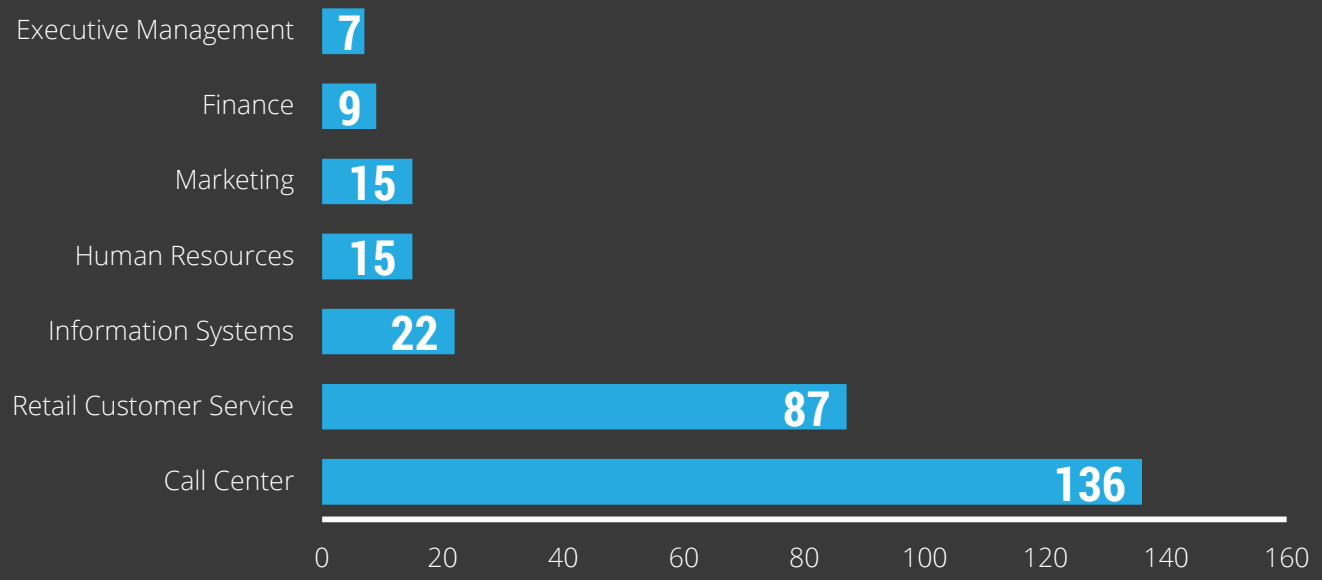


RANKING HORIZONTAL BAR CHARTS

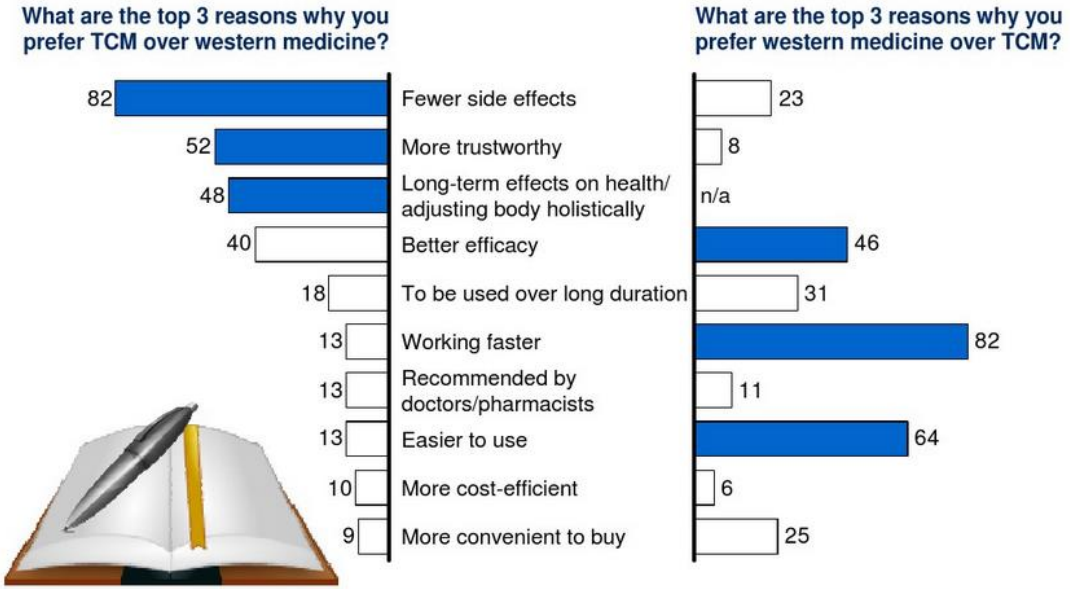
NUMBER OF EMPLOYEES BY FUNCTION

Full-time equivalents, January 2014



TCM is likely to see continued growth driven by high consumer acceptance …

Percent of respondents who chose specific reason as one of their top 3

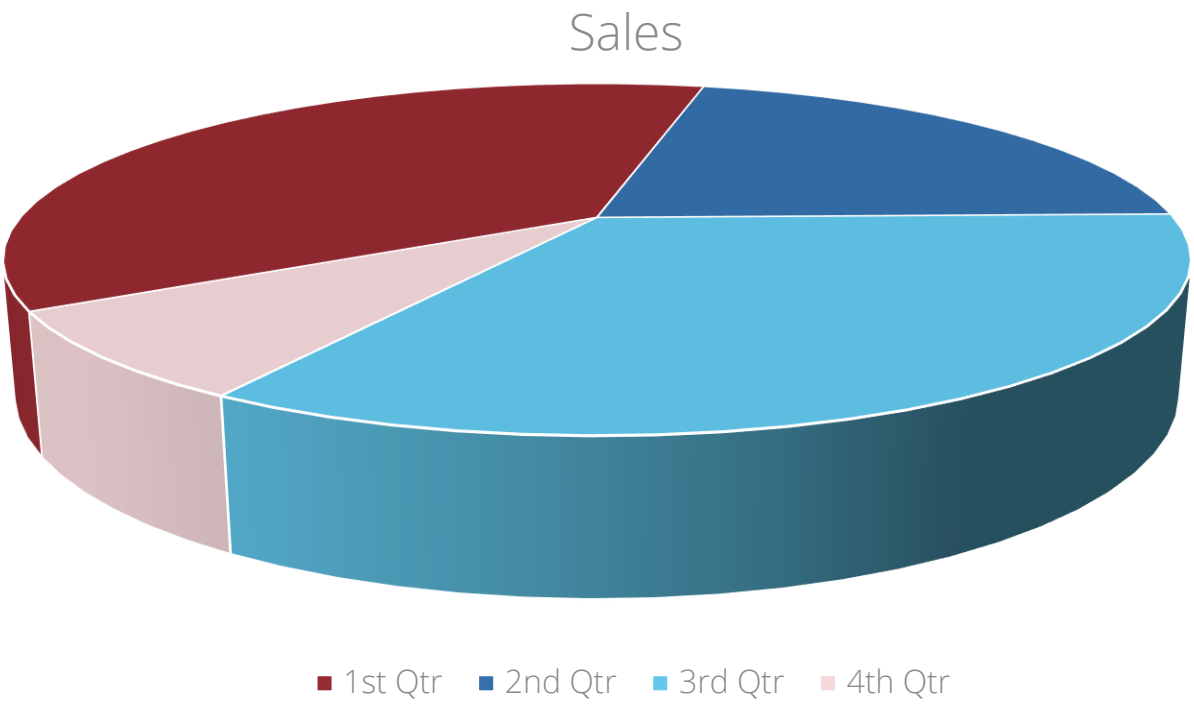


SOURCE: McKinsey China Consumer Center

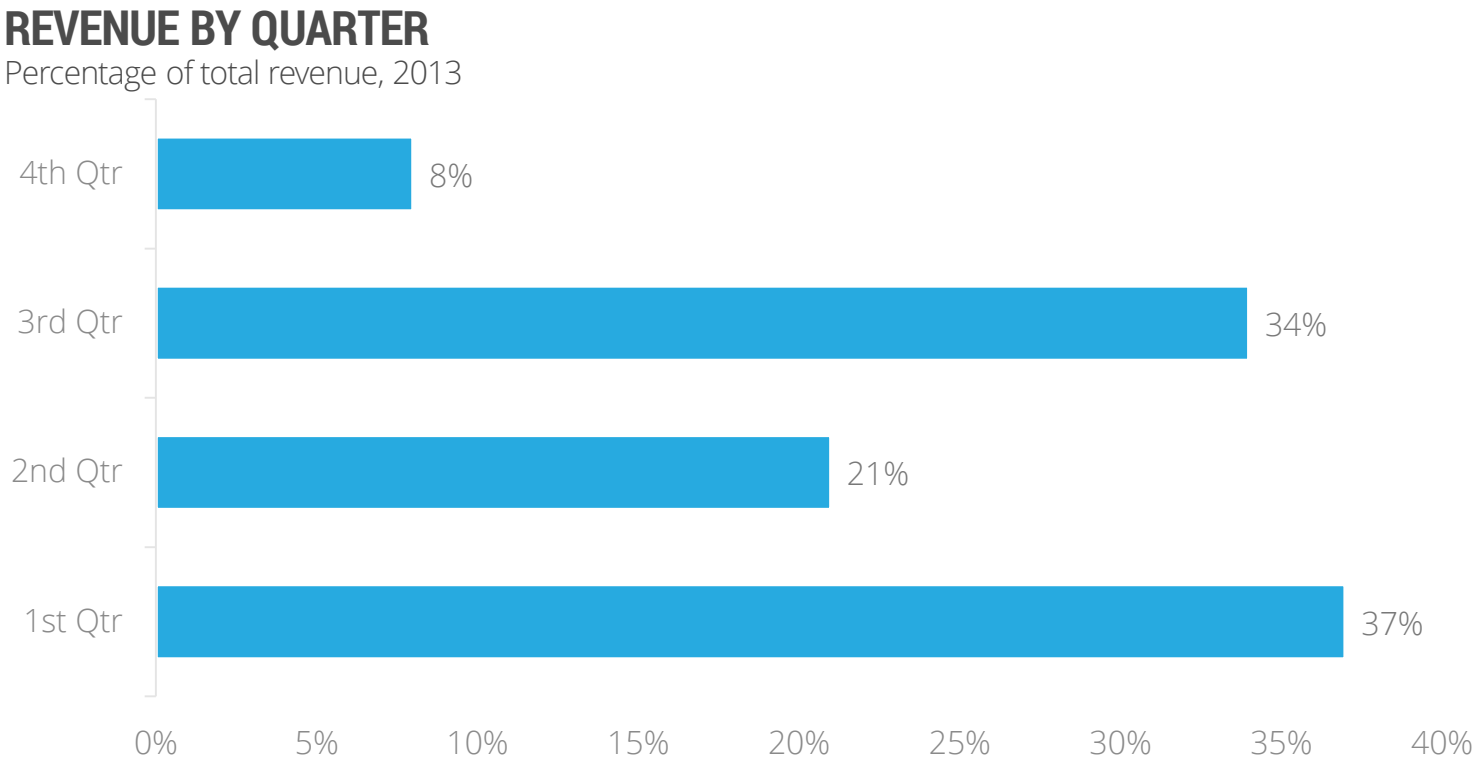
Basic Graphs

4. PART-TO-WHOLE

PIE CHARTS YUK



BAR CHARTS BETTER



GOLDEN RULE

DON'T USE PIE CHARTS

Our visual perception is not designed to accurately assign quantitative information to 2D areas. We have an even harder time when the third dimension of depth is added.

BAR CHARTS BETTER

REVENUE BY QUARTER

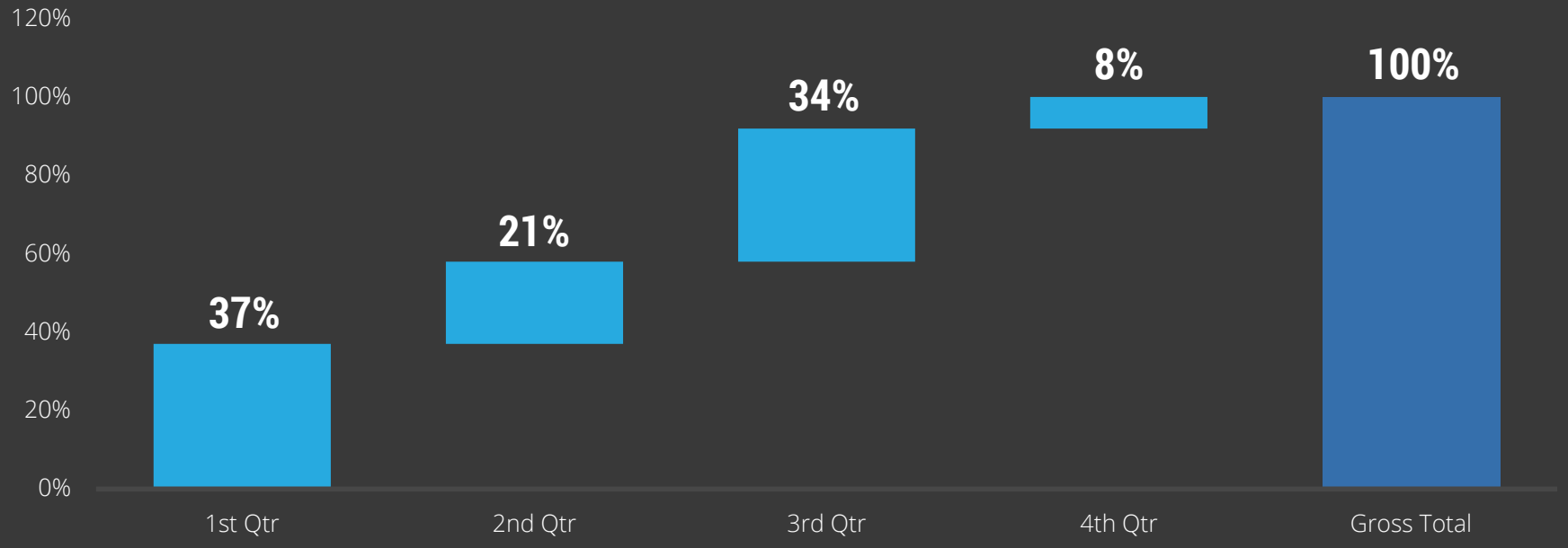
Percentage of total revenue, 2013

1st Qtr	2nd Qtr	3rd Qtr	4th Qtr		
37%	21%	34%	8%	=	100%

OPTIONAL WATERFALL CHART

REVENUE BY QUARTER

Percentage of total revenue, 2013



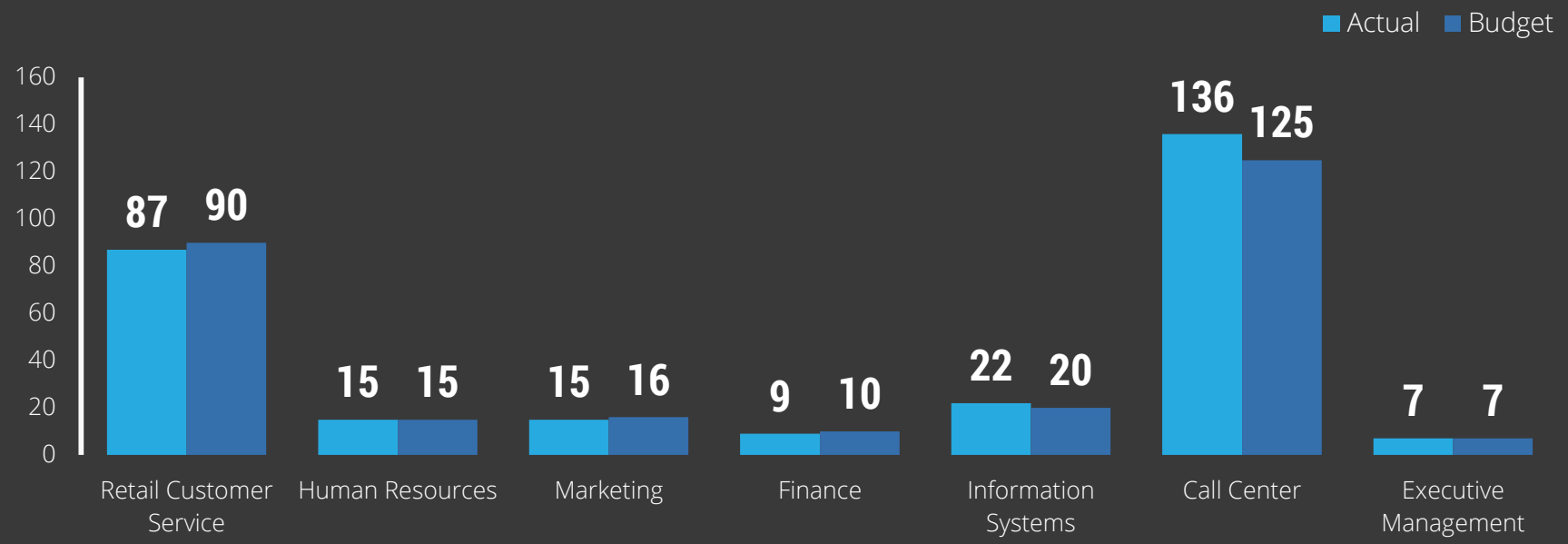
Basic Graphs

5. DEVIATION

DEVIATION HORIZONTAL BAR CHARTS

NUMBER OF EMPLOYEES BY FUNCTION, BUDGET VERSUS ACTUALS

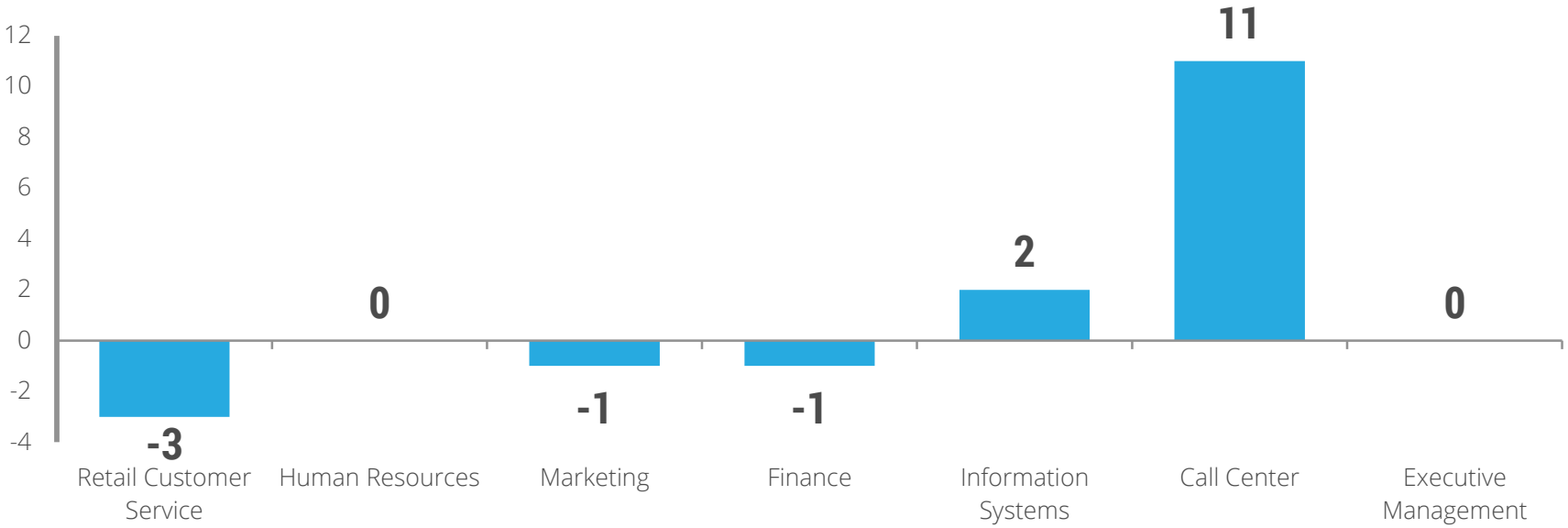
Full-time equivalents, January 2014



DEVIATION HORIZONTAL BAR CHARTS

NUMBER OF EMPLOYEES BY FUNCTION, VARIANCE TO BUDGET

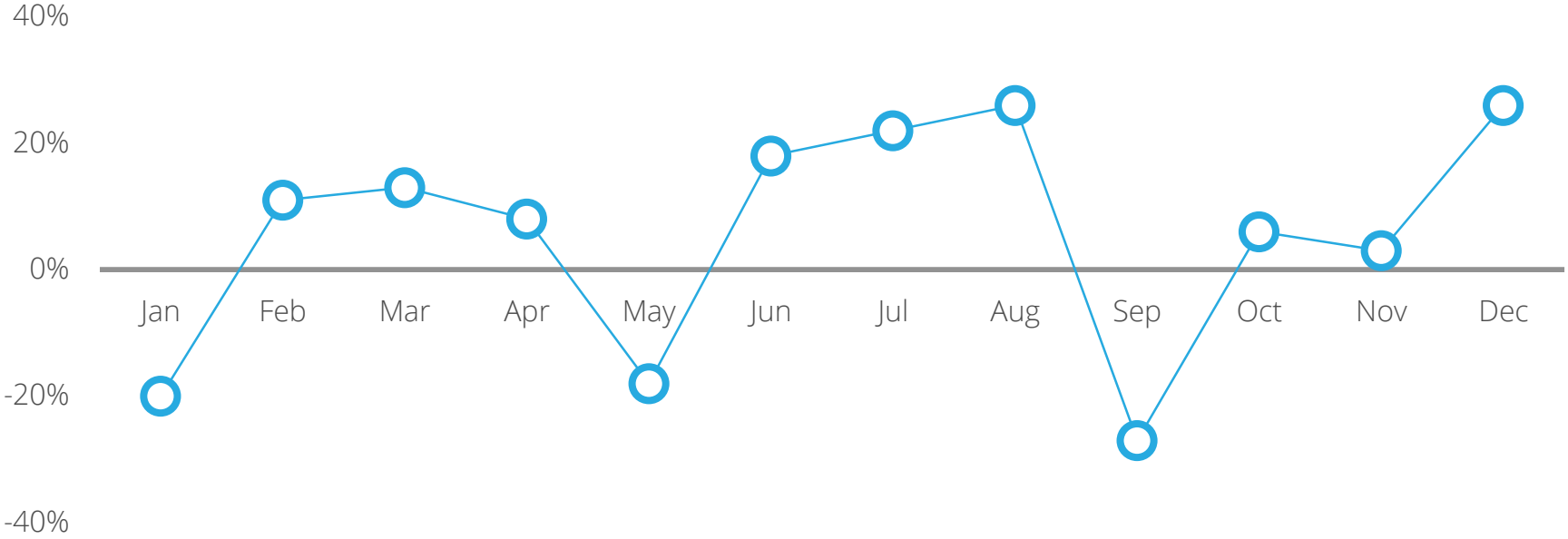
Full-time equivalents, January 2014



DEVIATION LINE GRAPHS

CHANGE IN REVENUE, BY MONTH

Percent, 2013



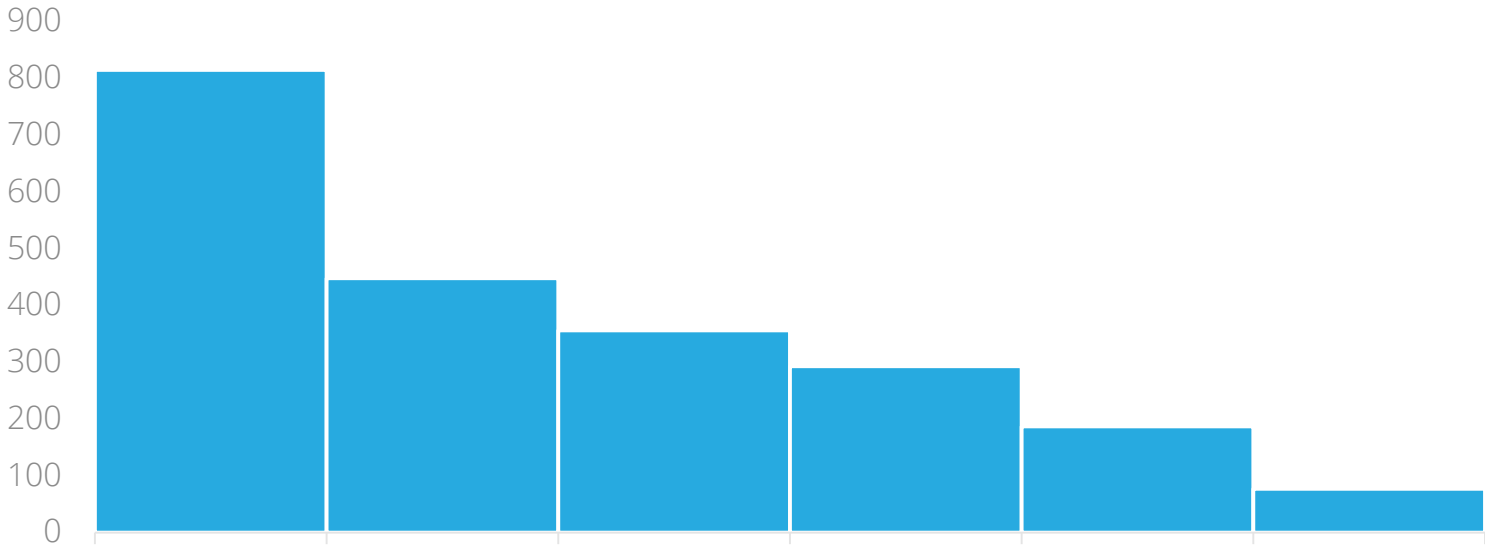
Basic Graphs

6. DISTRIBUTION

DISTRIBUTIONS HISTOGRAMS

NUMBER OF TRANSACTIONS, BY TRANSACTION SIZE

Number of transactions, 2013

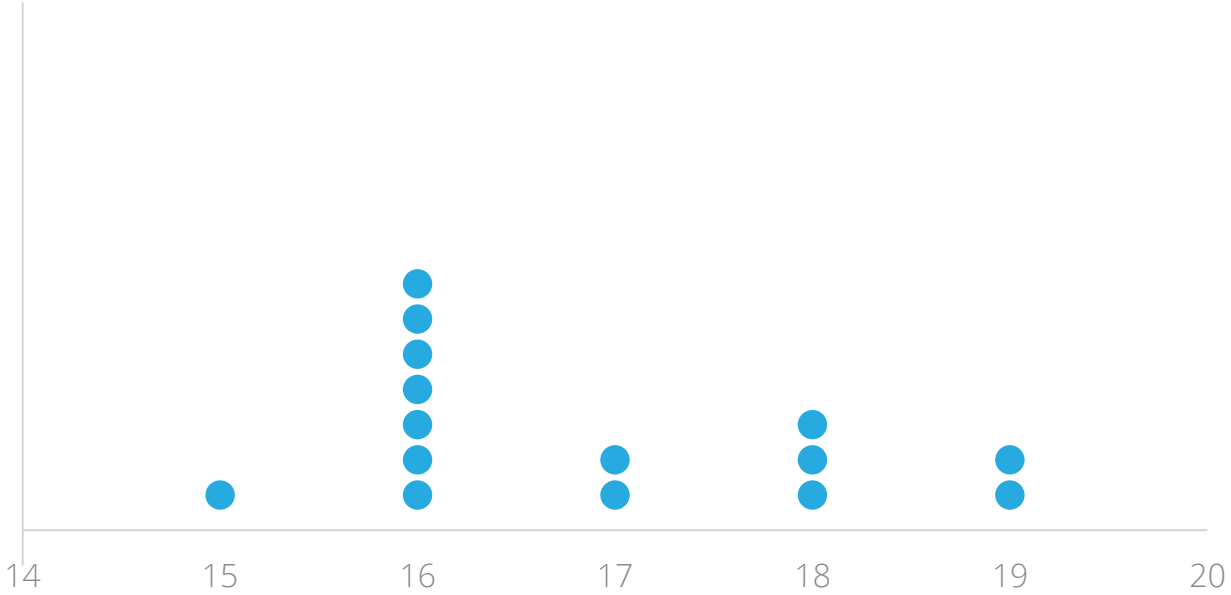


$0 to $10.00	$10.01 to $20.00	$20.01 to $30.00	$30.01 to $40.00	$40.01 to $50.00	>$50.00		
37%	21%	16%	13%	9%	4%	=	100%

DISTRIBUTIONS STRIP PLOT

NUMBER OF STUDENTS, BY AGE

As at 2013

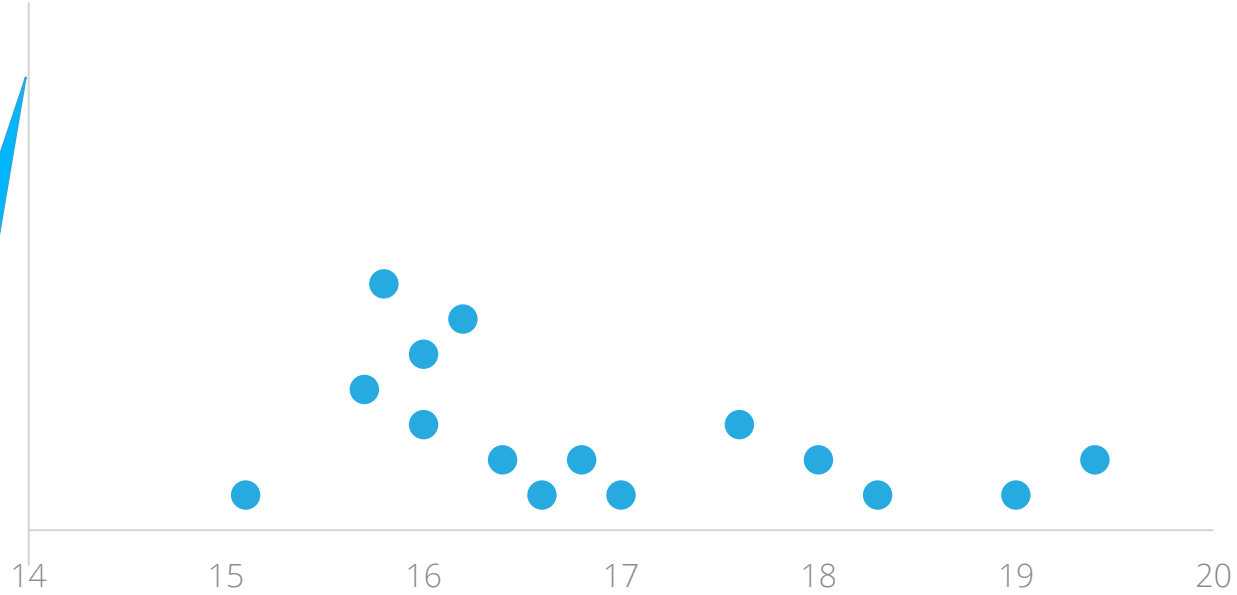


DISTRIBUTIONS JITTERED STRIP PLOT

NUMBER OF STUDENTS, BY AGE

As at 2013

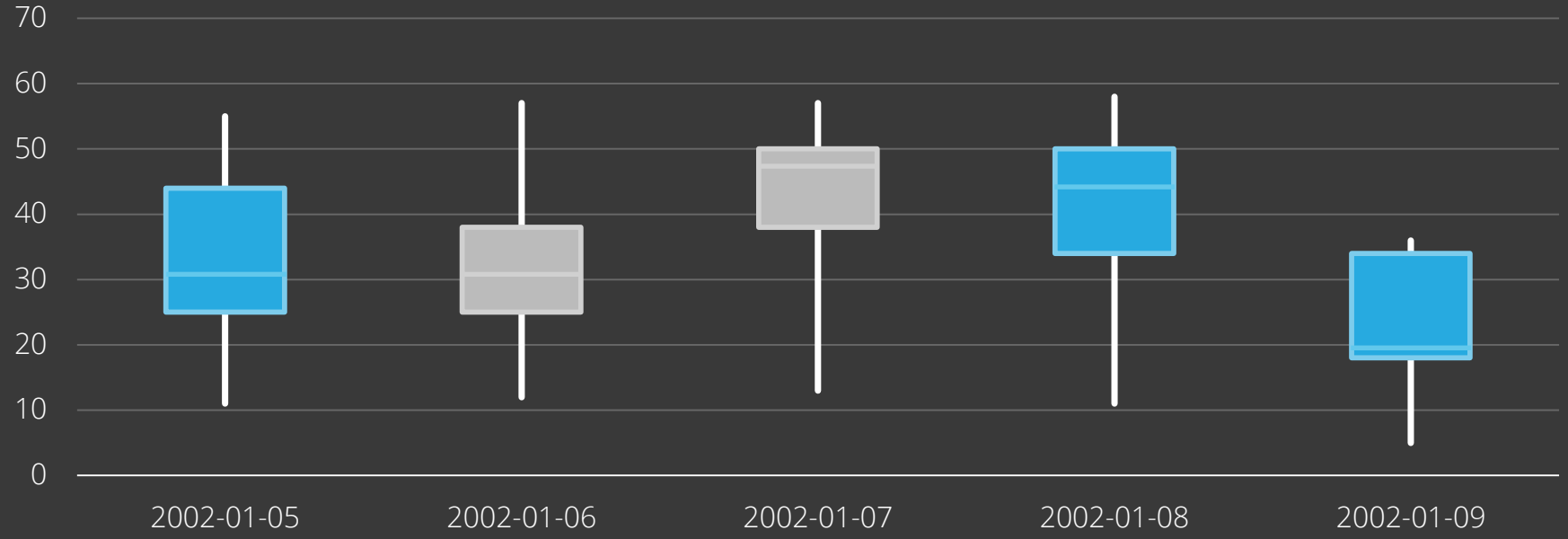
Y-axis is a random variable to separate observations



BOX PLOT CHART

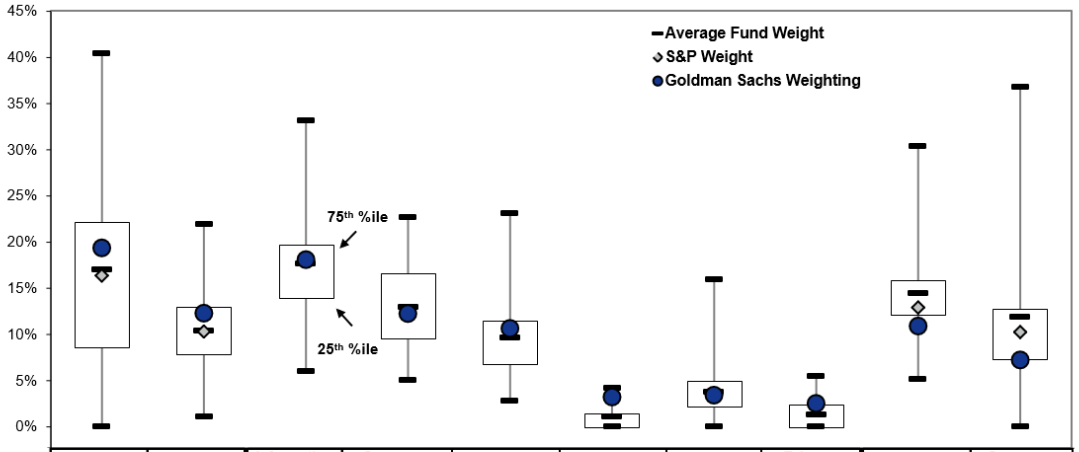
HIGH DEGREE OF VARIABILITY OF EMISSIONS

PPM, 2002,



Field test study conducted 2002.

Sector distribution of Lipper Large-Cap Core mutual funds based on equal-weighted average holdings



	Financials	Industrials	Information Technology	Consumer Discretionary	Energy	Materials	Utilities	Telecom Services	Health Care	Consumer Staples
S&P Weight	16.4	10.3	18.1	12.3	10.6	3.4	3.2	2.5	12.9	10.2
Average Mutual Fund Weight	17.0	10.4	17.6	12.9	9.6	3.8	1.1	1.3	14.4	11.9
Goldman Sachs Weighting	19.4	12.3	18.1	12.3	10.6	3.4	3.2	2.5	10.9	7.2
Goldman Sachs vs S&P 500 Overweight/(Underweight)	300 bp	200 bp	0 bp	0 bp	0 bp	0 bp	0 bp	0 bp	(200) bp	(300) bp
Percent difference OW/(UW)	18 %	19 %	0 %	0 %	0 %	0 %	0 %	0 %	(15)%	(29)%

+ ← Overweight / Underweight → –

Source: Lipper, Lionshares via FactSet and Goldman Sachs ECS Research. As of August 30, 2013.

Golman Sachs

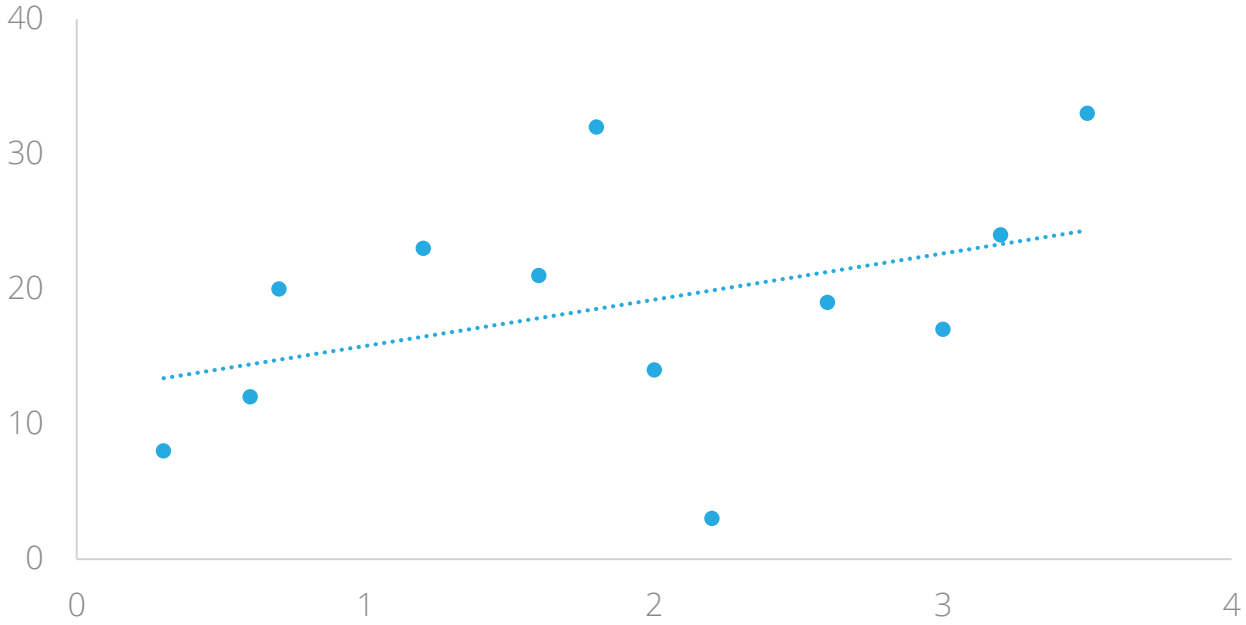
Basic Graphs

7. CORRELATION

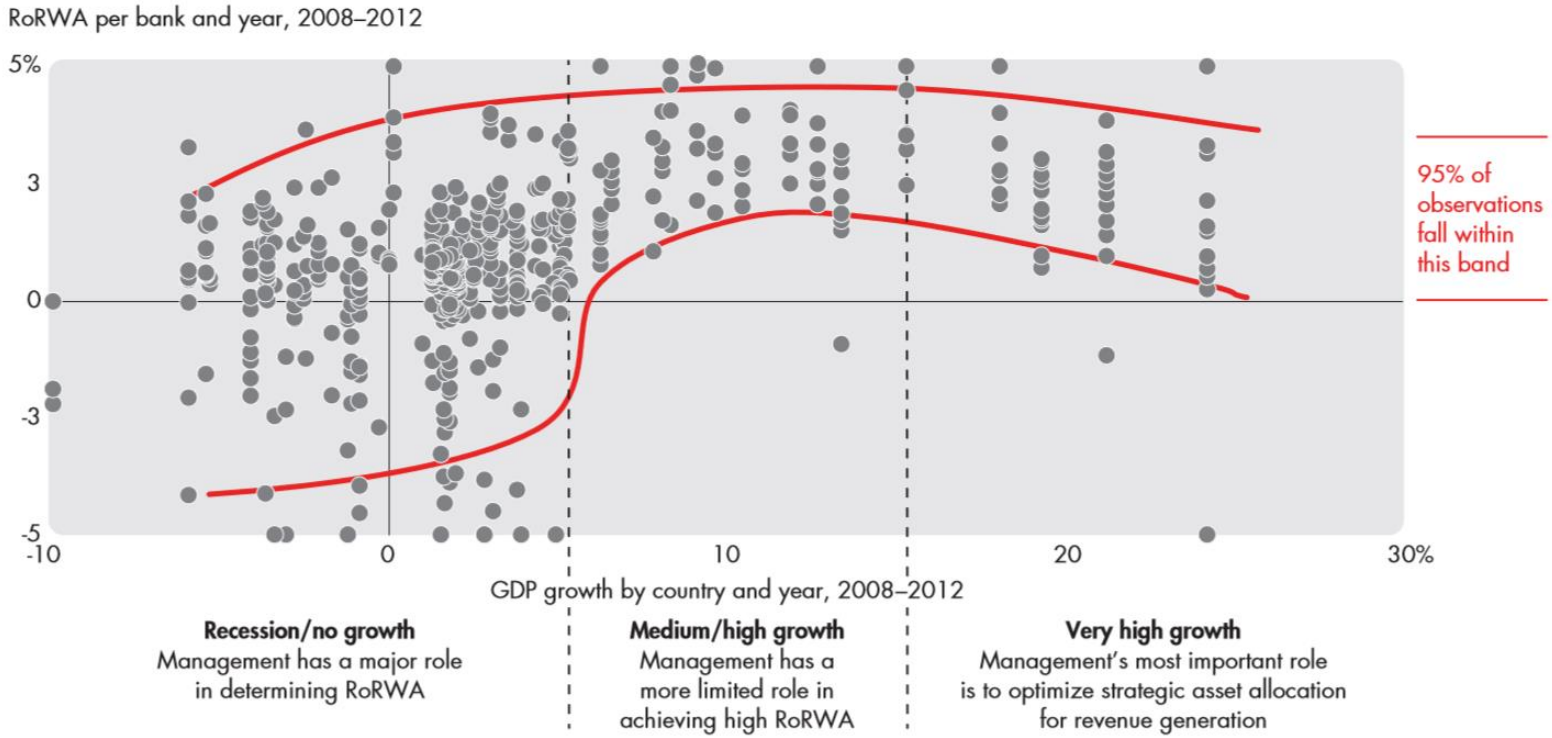
CORRELATIONS SCATTER PLOT

NUMBER OF TRIPS CORRELATED WITH AGE

2014

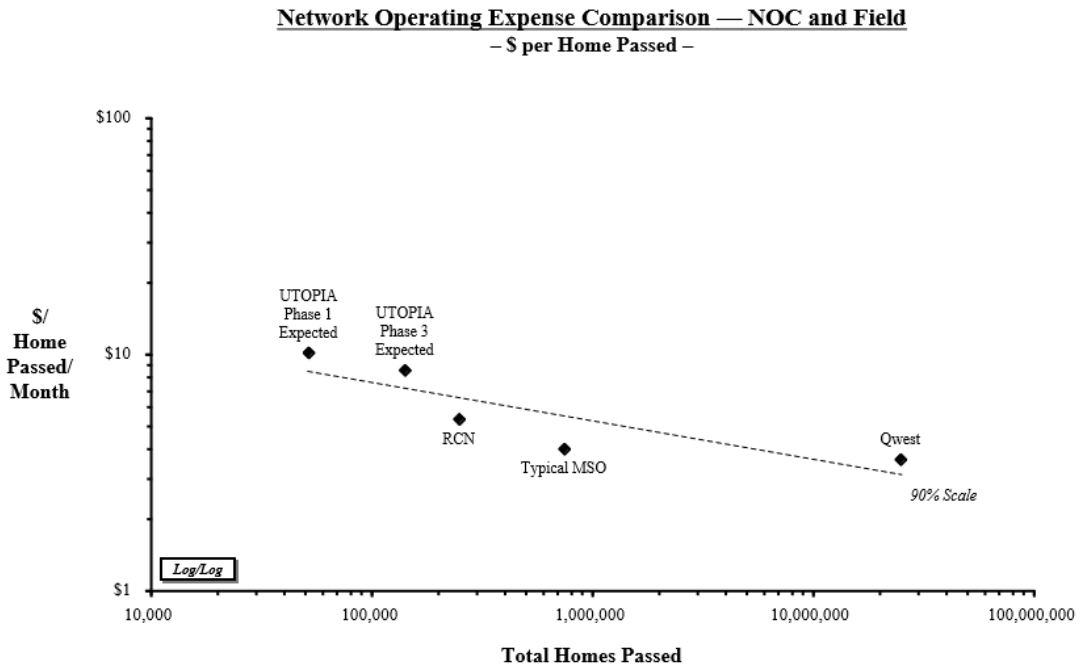


Although RoRWA correlates with economic growth, there is still substantial variation among banks



Note: Each observation is composed of a bank's RoRWA in a given year and the respective country's year-over-year GDP growth rate in that year.
Sources: Banks' financial statements 2008–2012, n=605; Bureau van Dijk; Bain analysis

UTOPIA's scale-adjusted projected network operations cost is comparable to other operators



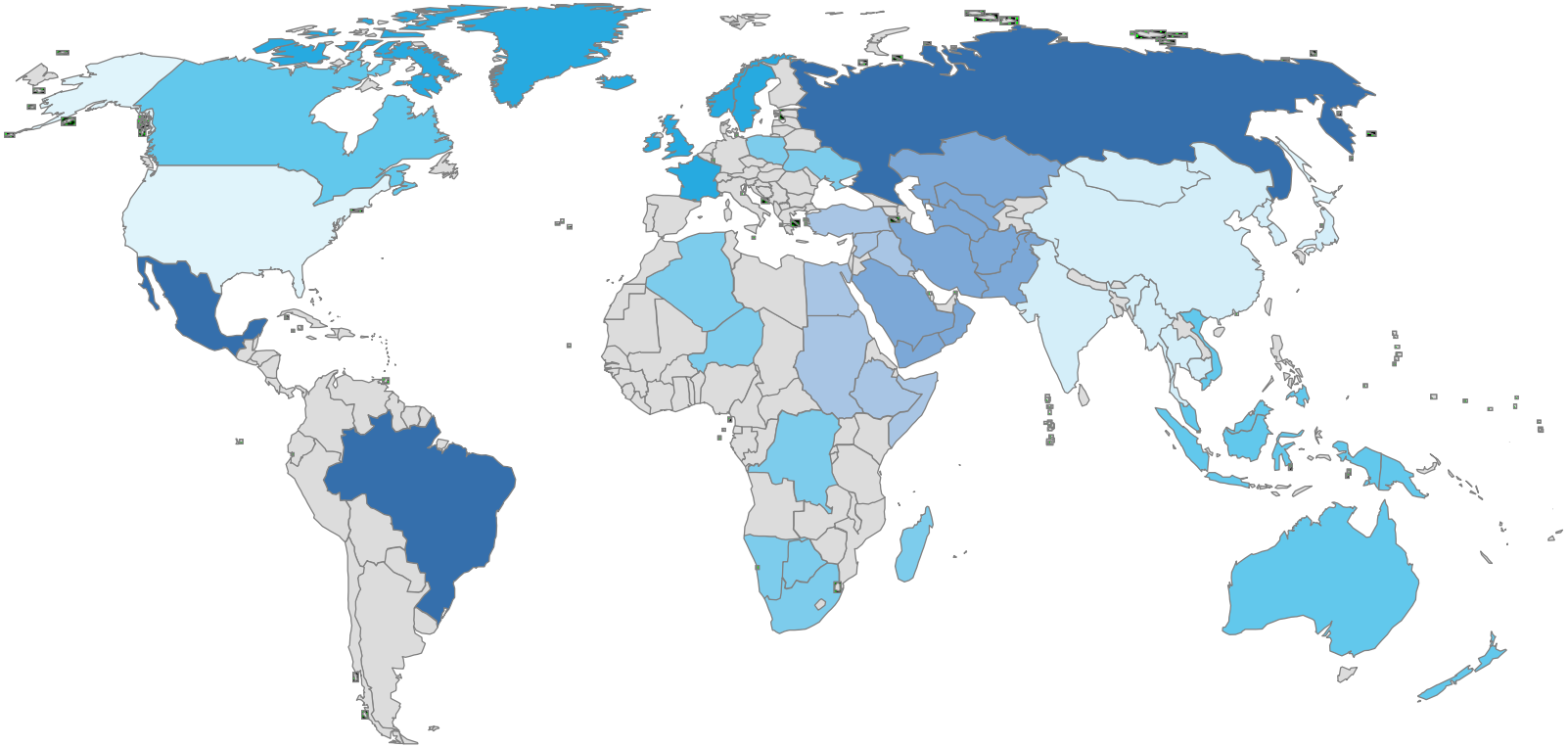
Note: NOC costs include asset management, headend, co-location, and interconnect expense. Field expenses include field maintenance and electronics maintenance.

Source: UTOPIA cost and revenue model, April 2004; company statements; Dean & Company research

Basic Graphs

8. GEOSPATIAL

GEOSPATIAL MAP



LESSON SUMMARY

NOMINAL COMPARISON
Use bar charts

TIME SERIES
Use line charts or bar charts

RANKING
Use bar charts

PART-TO-WHOLE
Use bar charts / histogram

DEVIATION
Use line charts

DISTRIBUTION
Use histograms or box-plots

CORRELATION
Use scatterplots

GEOSPATIAL
Use maps